

Aditya Kaleeswar

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EDUCATION

Vellore Institute of Technology

Sep 2022 – Jul 2026

B.Tech in Computer Science · CGPA 8.41 — Katpadi, Vellore

Maharishi Vidya Mandir

Apr 2018 – Jul 2022

Senior Secondary (CBSE) · 96% (XII), 94% (X) — Chetpet, Chennai

EXPERIENCE

Research Intern

Dec 2025 – Present

UTRGV (AIM Lab) × VIT Vellore · Remote / Vellore

- ▶ **Engineered Cementitious Composites (ECC)**: predicts the tensile strength and ductility of a concrete mix from its design parameters, so a mix can be tuned to a target performance at lower cost before anything is cast.
- ▶ Trains **CatBoost** quantile models (q10 / q50 / q90) calibrated with **Mondrian Conformal Quantile Regression**, giving a trustworthy 80% prediction interval for every mix instead of a single unverifiable number.
- ▶ **Physics-informed crack assessment**: scores how dangerous a structural crack is from one inspection photo. **YOLO-seg** and **SAM 2** segment each crack, **Depth Anything V2** with mm-per-pixel calibration recovers metric scale, and branch-splitting + fragment-linking measure its length, angle and width.
- ▶ A **physics-informed neural network** then solves the linear-elastic-fracture-mechanics Mode-I stress-intensity factor (K_1) per crack — trained once per geometry ratio, microseconds at inference — and ranks severity as K_1/K_{IC} against the material's fracture toughness.

[ECC preprint](#) · [ECC repo](#) · [PINN preprint](#) · [PINN repo](#)

Research Intern

Oct 2024 – Jun 2025

Samsung Research, Bangalore · Remote

- ▶ Built **BQ2Viz**, a multi-agent GenAI framework (LangGraph · LangChain · ChromaDB) that turns natural language into data visualizations via a profiling agent + a self-correcting code generator.
- ▶ Designed a **reasoning-based planner** for multi-file retrieval, inferring multiple distinct charts from a single abstract query.
- ▶ Benchmarked vs **NVAGENT** on **VisEval**: **84.57%** accuracy on complex multi-table queries; cut error rate to **11.5%** with a feedback-driven repair loop.

[NL→Data-Viz repo](#)

Software Engineer — Autonomous Domain (Senior Autonomous Member)

Apr 2024 – Present

Team Vyadh (Mars Rover Team), VIT · Vellore

- ▶ Built the autonomous navigation stack (**ROS 2 / Nav2 + SLAM**) with a **Frontier Exploration** algorithm to map unknown environments within dynamic GPS geofences.
- ▶ **C++-based obstacle & pit avoidance** using **Intel RealSense D455 + PCL**, with odometry path recovery and a PID alignment to compensate for wheel drift.
- ▶ Fused IMU + odometry with an **Extended Kalman Filter** to cut state-estimation noise; built a telemetry GUI for Astro-Bio streams (CO₂, IR, moisture). Team placed **13th at IRC** and **5th at IRDC**.

[Obstacle-avoidance repo](#) · [Rover-stack repo](#)

PROJECTS

HazMap — Autonomous Multi-Hazard Mapping Rover

Dec 2025 – May 2026

ROS 2 · NVIDIA Jetson Orin Nano · EKF · GADEN · Bayesian Estimation · Python

- ▶ Autonomous rover that builds **chemo-visual hazard maps** of unknown environments, fusing LiDAR and gas-sensor data with an **Extended Kalman Filter**.
- ▶ **Coverage planning**: a modified **C*** algorithm incrementally builds a **Reachability Connectivity Graph (RCG)** and drives boustrophedon laps until the full navigable area is covered.
- ▶ **Gas-source localization (ROS 2, validated in GADEN)**: **CUSUM**-based hotspot detection, gas-concentration mapping, and **Bayesian** probabilistic source estimation to pinpoint leak origins.

preprint · hazpatrol repo · CGLS_gaden repo

Alpha Stack — Natural Language → Production Codebases

Aug 2025 – Nov 2025

Python · Multi-Agent Orchestration · RAG · NetworkX (DAGs) · Docker (sandbox) · CLI

- ▶ Hierarchical multi-agent system (**Dominant–Worker** model) that turns a natural-language prompt into a complete, production-ready, multi-file codebase.
- ▶ **Tree-based DFS traversal** propagates semantic context across files; an incremental **Dependency Analysis Engine** on Directed Acyclic Graphs (NetworkX) maps inter-file relationships in $O(k)$ time, preventing circular dependencies.
- ▶ A self-healing **test → fail → diagnose → repair** loop drives validation: **79.10%** executability on the ProjectDev benchmark vs ChatDev’s 32.79%.

preprint · repo

PATENT

Force Sensor-Based Train Presence & State Detection

Indian Patent Application No. 202641007500

ESP32 · Force-Sensitive Resistors (FSR) · Embedded C · Finite-State Machine · IoT

- ▶ Platform-level IoT system that **passively** detects and classifies a train’s state — approaching, stopped, pass-through, departing — and its **direction of entry**, with no train-side equipment.
- ▶ An array of **Force-Sensitive Resistors** under the track measures the train’s normal force; distributed **ESP32** nodes do edge thresholding and send digital triggers to a central ESP32 running a **finite-state machine** that classifies state from trigger sequence and timing.
- ▶ **Minimally invasive and low-cost** (no track-circuit modifications), enabling platform automation such as automatic barriers, alerts, and announcements.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript (ES6+), SQL (Postgres), HTML/CSS
GenAI & ML: PyTorch, LangGraph, LangChain, RAG, Pinecone, ChromaDB, Ollama, OpenRouter, CatBoost, Conformal Prediction
Robotics & Vision: ROS 2 (Nav2), SLAM, OpenCV, PCL, YOLOv8, TensorRT, NVIDIA Jetson, ESP32
Full Stack & DB: React, Node.js, Express, MongoDB, MERN, REST APIs, Vite
Data & Algorithms: pandas, NumPy, NetworkX, SciPy, Matplotlib, scikit-learn, DuckDB
DevOps & Cloud: GCP, Docker, Git, Linux